WEEK 2 (Day 1 & Day 2)

Lists in Python

A list is an ordered, mutable (changeable) collection that can store multiple items of different data types. Lists are created using square brackets [].

```
fruits = ["apple", "banana", "cherry", "kiwi", "plum", "orange"]
print(fruits)
print(type(fruits))
```

```
C:\Users\bhavn\OneDrive\Desktop\DataAn1\.venv\Scripts\python.exe
['apple', 'banana', 'cherry', 'kiwi', 'plum', 'orange']
<class 'list'>

Process finished with exit code 0
```

Common List Operations

1. Accessing Elements
2. Adding Elements
3. Inserting an Element
4. Removing Elements
5. Updating an Element
6. Length of a List
7. Traversing a List

```
fruits = ["apple", "banana", "cherry", "kiwi", "plum", "orange", "cherry"]
print(fruits)
print("Length of list :",len(fruits))  # length
print("No. of times cheery occurs :",fruits.count("cherry")) #count the occurrence of an entity
print("No. of times kiwi occurs :",fruits.count("kiwi"))
fruits.append("watermelon") # adding contents to the list (adds at the last position)
print(fruits)
fruits.insert( index: 3, object: "apricot") # inserts an entity at a specific location
print(fruits)
fruits.remove("cherry") # removes the specified entity
print(fruits)
fruits.pop(5) # removes or pops an element from the specified location
print(fruits)
```

```
C:\Users\bhavn\OneDrive\Desktop\DataAn1\.venv\Scripts\python.exe C:\Users\bhavn\OneDrive\Desktop\DataAn1\lists.py
['apple', 'banana', 'cherry', 'kiwi', 'plum', 'orange']
<class 'list'>
['apple', 'banana', 'cherry', 'kiwi', 'plum', 'orange', 'cherry']
Length of list : 7
No. of times cheery occurs : 2
No. of times kiwi occurs : 1
['apple', 'banana', 'cherry', 'kiwi', 'plum', 'orange', 'cherry', 'watermelon']
['apple', 'banana', 'cherry', 'apricot', 'kiwi', 'plum', 'orange', 'cherry', 'watermelon']
['apple', 'banana', 'apricot', 'kiwi', 'plum', 'orange', 'cherry', 'watermelon']
['apple', 'banana', 'apricot', 'kiwi', 'plum', 'cherry', 'watermelon']

Process finished with exit code 0
```

```python
# list comprehension
l1 = [10,20,30,40,50]
l2 = []
for i in l1:
    if(i>25):
        l2.append(i)

print(l2)
print(l1)
# another method to do the same
l3 = [i for i in l1]
print(l3)
```

```
C:\Users\bhavn\OneDrive\Desktop\DataAn1\.venv\Scripts\python.exe
['apple', 'banana', 'cherry', 'kiwi', 'plum', 'orange']
<class 'list'>
[30, 40, 50]
[10, 20, 30, 40, 50]
[10, 20, 30, 40, 50]

Process finished with exit code 0
```

```python
#Iteration using for loop
fruits = ["apple", "banana", "cherry", "kiwi", "plum", "orange"]
for i in fruits:
    print(i)
```

```
C:\Users\bhavn\OneDrive\Desktop\DataAn1\.venv\Scripts\python.exe
apple
banana
cherry
kiwi
plum
orange

Process finished with exit code 0
```

Tuples in Python

A **tuple** is an ordered, immutable (unchangeable) collection of items. Tuples are created using parentheses ().

```
a = "apple" , "mango" , "cherry"
print(a)
print(type(a))

# if our tuple contains only 1 value then end it with a coma , otherwise it's type is by default assumed to be string
b = "red"
print(b)
print(type(b))
c = "blue" ,
print(c)
print(type(c))
```

```
C:\Users\bhavn\OneDrive\Desktop\DataAn1\.venv\Scripts\python.exe
('apple', 'mango', 'cherry')
<class 'tuple'>
red
<class 'str'>
('blue',)
<class 'tuple'>

Process finished with exit code 0
```

Common Tuple Operations

1. Accessing Elements
2. Tuple Slicing
3. Finding Length
4. Traversing a Tuple
5. Counting an Element
6. Finding Index of an Element

```
a = ("oneplus", "vivo", "redmi", "iphone", "nokia", "oppo")
print(a)
print(a[2:5])
print(a[:4])
print(a[::2])
print(a[::-1])
print(a[2::-1])
```

```
C:\Users\bhavn\OneDrive\Desktop\DataAn1\.venv\Scripts\python.exe
('oneplus', 'vivo', 'redmi', 'iphone', 'nokia', 'oppo')
('redmi', 'iphone', 'nokia')
('oneplus', 'vivo', 'redmi', 'iphone')
('oneplus', 'redmi', 'nokia')
('oppo', 'nokia', 'iphone', 'redmi', 'vivo', 'oneplus')
('redmi', 'vivo', 'oneplus')

Process finished with exit code 0
```

```python
# # with for loop
a = ("oneplus", "vivo", "redmi", "iphone", "nokia", "oppo")
for i in a:
    print(i)
print("\n")
```

```
C:\Users\bhavn\OneDrive\Desktop\DataAn1\.venv\Scripts\python.exe
oneplus
vivo
redmi
iphone
nokia
oppo
```

```python
# conversion of tuples and tuple functions
a = ("oneplus", "vivo", "redmi", "iphone", "nokia", "oppo")
print(a)
print("Before conversion")
print(type(a))
a = list(a)
print("After conversion")
print(type(a))
a.append("Nothing")
print(a)
a = tuple(a)
print(type(a))
print("No. of times nokia occurs :",a.count("nokia"))
print("nokia occurs at index position :",a.index("nokia"))
```

```
C:\Users\bhavn\OneDrive\Desktop\DataAn1\.venv\Scripts\python.exe C:\
('oneplus', 'vivo', 'redmi', 'iphone', 'nokia', 'oppo')
Before conversion
<class 'tuple'>
After conversion
<class 'list'>
['oneplus', 'vivo', 'redmi', 'iphone', 'nokia', 'oppo', 'Nothing']
<class 'tuple'>
No. of times nokia occurs : 1
nokia occurs at index position : 4

Process finished with exit code 0
```

Dictionaries in Python

A **dictionary** is an **unordered, mutable (changeable)** collection of data stored as **key-value pairs**. Each key in a dictionary must be unique. Dictionaries are created using **curly braces {}**.

```
Employee_Data = {"name" : "John" , "age" : 25, "gender":"male"}
print(Employee_Data)
print(Employee_Data["name"])
```

```
C:\Users\bhavn\OneDrive\Desktop\DataAn1\.venv\Scripts\python.exe
{'name': 'John', 'age': 25, 'gender': 'male'}
John

Process finished with exit code 0
```

Common Dictionary Operations

1. Accessing Values
2. Adding a New Key-Value Pair
3. Updating a Value
4. Removing an Item
5. Finding the Length
6. Displaying Keys
7. Displaying Values
8. Displaying Key-Value Pairs
9. Traversing a Dictionary

```python
#for printing key names oe by one
Student = {"name" : "Isabella", "class" : "8th", "RollNo." : 23}
for x in Student:
    print(x)
```

```
C:\Users\bhavn\OneDrive\Desktop\DataAn1\.venv\Scr
name
class
RollNo.

Process finished with exit code 0
```

```python
# printing all the value names one by one
Student = {"name" : "Isabella", "class" : "8th", "RollNo." : 23}
for x in Student:
    print(Student[x])
```

```
C:\Users\bhavn\OneDrive\Desktop\DataAn1\.venv\Scripts\python.exe
Isabella
8th
23

Process finished with exit code 0
```

```python
# using item function
Student = {"name" : "Isabella", "class" : "8th", "RollNo." : 23}
for x , y in Student.items():
    print(x,"-",y)
```

```
C:\Users\bhavn\OneDrive\Desktop\DataAn1\.venv\Scripts\pyth
name - Isabella
class - 8th
RollNo. - 23

Process finished with exit code 0
```

```python
# nested dictionaries
Employees = {1:{"Name":"John", "Age":34,"Gender":"Male"},
             2:{"Name":"Sia", "Age":27,"Gender":"Female"},
             3:{"Name":"George", "Age":65,"Gender":"Male"}}
print(Employees)
print(Employees[3]["Name"])
```

```
C:\Users\bhavn\OneDrive\Desktop\DataAn1\.venv\Scripts\python.exe C:\Users\bhavn\OneDrive\Desktop\DataAn1\dictionaries.py
{1: {'Name': 'John', 'Age': 34, 'Gender': 'Male'}, 2: {'Name': 'Sia', 'Age': 27, 'Gender': 'Female'}, 3: {'Name': 'George', 'Age': 65, 'Gender': 'Male'}}
George

Process finished with exit code 0
```